

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		This is the energy needed to eject an electron (1) from the surface or the metal or rubidium (not the atom) (1)	2			2		
	(b)		Any electron ejected is ejected by a single photon or equivalent (1) Photon energy [or hf] must be [at least] equal to ϕ (1) So $hf > \phi$ [for emission] (1)	1 1	1		3		
	(c)	(i)	Mean = 0.377 [V] Accept 0.376 [V] or 0.3765 [V] or 0.38 (V) (1) ± 0.011 [V] or 0.0110 [V] or 0.01 [V] (1) Answer + uncertainty expressed to an appropriate number of sig figs i.e. 0.377 (or 0.376) V $\pm$ 0.011 V or 0.38 ± 0.01 [V] (1) Must use earlier calculations		3			3	3
		(ii)	$E_{k \max}$ calculated for one stopping pd i.e. $0.366 \text{ V} \rightarrow 5.86 \times 10^{-20} \text{ J}$, or $0.388 \text{ V} \rightarrow 6.21 \times 10^{-20} \text{ J}$ or ($0.377 \text{ V} \rightarrow 6.03 \times 10^{-20} \text{ J}$) (1) Einstein's equation used correctly, either with hf and one $E_{k \max}$ value including mean to give ϕ , or with hf and $\phi = 3.62 \times 10^{-19} \text{ J}$ to give $E_{k \max}$ e.g. $\phi = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 5.86 \times 10^{-20} \text{ (1)}$ $= 3.62 \times 10^{-19} \text{ [J] (1)}$ or $\phi = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 6.21 \times 10^{-20} \text{ (1)}$ $= 3.58 \times 10^{-19} \text{ [J] (1)}$ or $E_{k \max} = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 3.62 \times 10^{-19} \text{ (1)}$ $= 5.83 \times 10^{-20} \text{ [J] (1)}$						

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			Conclusion: the ϕ of Rb, (3.62×10^{-19} J) just fits in the range 3.58×10^{-19} J to 3.62×10^{-19} J so Rb is a possibility or $E_{k\max} = 5.83 \times 10^{-20}$ J and lies just outside the range 5.86×10^{-20} J to 6.21×10^{-20} J) so possibly Rb or can't be Rb (1) Variation on this method for the 1st and 4th marks: 5.83×10^{-20} J correspond to a stopping pd of 0.364 V (1) which lies just outside the range given, so possibly / probably not Rb (1)			4	4	2	4
			Question 6 total	4	4	4	12	5	7